Muon-Collider $ZH \to 4b$ Cross Sections

Unpolarized $\mu^+\mu^-$ collisions at $\sqrt{s} = 1$ and 2 TeV

Revised selection: common dijet mass windows of 80–130 GeV

Explicit process and simulated decays

$$\boxed{\mu^+\mu^- \longrightarrow ZH \longrightarrow (Z \to b\bar{b})\,(H \to b\bar{b})}$$

The LO matrix element was generated in MadGraph5_aMC@NLO v3.6.4 with the resonant decay-chain definition

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generate mu+ mu- > z h QCD=0 QED=2, z > b b~, h > b b~
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Beam configurations and generation statistics

Quantity	$\sqrt{s} = 1$ **TeV**	$\sqrt{s} = 2$ **TeV**
Incoming μ^+ energy	500 GeV	1000 GeV
Incoming μ^- energy	500 GeV	1000 GeV
Beam polarization (P_{μ^+}, P_{μ^-})	$(0\%, 0\%)$	$(0\%, 0\%)$
Generated events / requested events	30 000/30 000	30 000/30 000

LO cross sections and fiducial acceptance

Quantity	$\sqrt{s} = 1$ **TeV**	$\sqrt{s} = 2$ **TeV**
$\sigma(\mu^+\mu^- \to ZH \to 4b)$, before fiducial cuts	1.564500 fb	0.378880 fb
Accepted events after fiducial cuts	29 284	29 241
Fiducial acceptance	97.6133%	97.4700%
$\sigma_{\text{fid}}(ZH \to 4b)$	(1.527161 ± 0.001589) fb	(0.369294 ± 0.000408) fb

Common fiducial selection

$$|\eta_b| < 5 \quad \text{for each of the four bottom quarks,} \qquad p_T^{\min}(b) = 0,$$
$$80\,\text{GeV} < m_{b\bar{b}}^{(1)} < 130\,\text{GeV},$$
$$80\,\text{GeV} < m_{b\bar{b}}^{(2)} < 130\,\text{GeV}.$$

The selection requires two *disjoint, opposite-sign* $b\bar{b}$ pairs, each within the common 80–130 GeV window; either pairing assignment is accepted. These revised windows follow the requested wider selection and supersede the previous Z (80–100 GeV) and Higgs (110–130 GeV) windows.

Normalization and accuracy notes

The "before cuts" rates already include both $Z \to b\bar{b}$ and $H \to b\bar{b}$ decays; *no additional branching-ratio factors* have been applied. Quoted fiducial errors are approximate quadrature combinations of selection-statistics and MadGraph integration MC errors (respectively 0.001379 and 0.000790 fb at 1 TeV; 0.000344 and 0.000221 fb at 2 TeV). The uncertainties exclude covariance, theoretical and detector systematics. An earlier independent event sample gives compatible estimates, but dedicated precision and convergence studies remain outstanding.